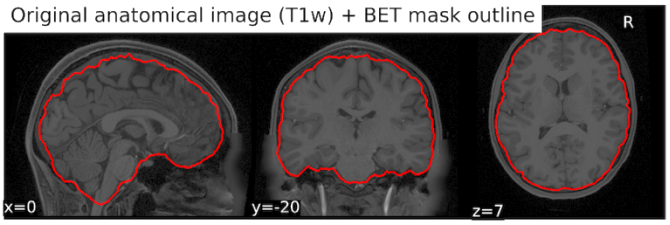
NX-421 Miniproject 1 – Report

Group T | **Variant 3**: K-means
Marbach Valentin, Schaffhauser Kilian, Martignoni Matteo

This project seeks to analyse fMRI data from a language task, taken from the Human Connectome Project (HCP), which was adapted from an experiment run by Binder and colleagues (Binder *et al.*, 2011) [1], by mapping language and semantic processing against mathematical processing with a hypothesis-driven Generalised Linear Model (GLM; Part 1) followed by a data-driven Multi-voxel Pattern Analysis (K-means; Part 2). The experiment alternates auditory story and math blocks, totalling a 3.8min run. The dataset contains a T1-weighted anatomical image, a task fMRI (LR run) and the event onset times.



1. Preprocessing and GLM

1.1 Structural preprocessing

Skull-stripping. This removes skull, scalp and neck, which carry no brain signal and disturb segmentation and registration. We used FSL Brain Extraction Tool (BET) with robust centre estimation and a fractional intensity threshold of 0.3. A stricter threshold of 0.5 (Annex 1) removes some brain tissue, so we kept the conservative 0.3 at the cost of a little remaining non-brain tissue (e.g., orbit/sinus region).

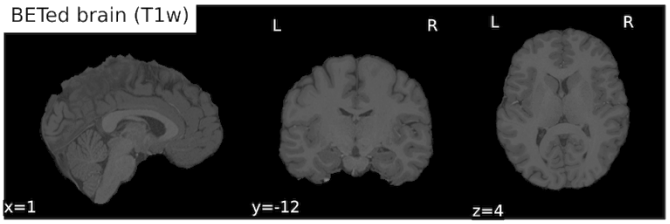


Figure 1 T1w with BET mask overlay on sagittal, coronal and axial slices. The extracted brain is shown in the second subplot.

Segmentation. This separates tissues to check the anatomy and to provide tissue masks (e.g., grey matter, where the BOLD signal originates). We used FSL FAST to obtain grey matter, white matter and CSF (volumes: CSF 229 mL, GM 654 mL, WM 496 mL; Figure 2).

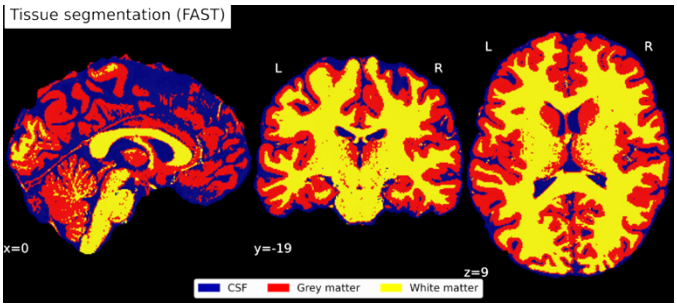


Figure 2 Tissue segmentation. Segmented tissues are overlaid on the BETed brain.

Registration to MNI. This brings the data into a common, standard space so that activations can be compared with atlases and literature coordinates. We compared two transformations (Figure 3). FLIRT, affine with 12 DOF. The scale factors (1.08/1.12/1.18) show that the subject's brain is smaller than the template, which a rigid 6-DOF transform could not absorb. The correlation with the MNI template is 0.854. ANTs SyN non-linear registration gives the best overlap with the template (dice 0.940 vs 0.894 with FLIRT) notably at the ventricles and sulci, but has a 0.809 correlation with MNI. We kept it, because the HCP fMRI is itself non-linearly normalised to MNI, so the T1 must match it equally well. The same transform was applied to the segmentation to yield the grey matter mask in MNI space, used for the GLM.

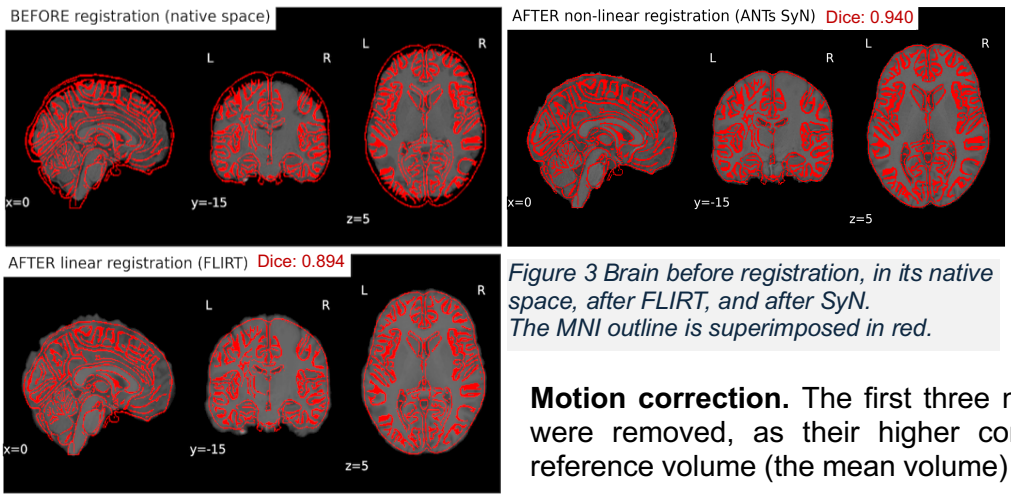


Figure 3 Brain before registration, in its native space, after FLIRT, and after SyN. The MNI outline is superimposed in red.

1.2 Functional preprocessing

Motion correction. The first three non-stationary volumes were removed, as their higher contrast would bias the reference volume (the mean volume) of the realignment. We

then applied motion correction (MCFLIRT) so that each voxel follows the same brain location over time. Its 6 parameters are then used as nuisance regressors in the GLM. The framewise displacement (FD; Figure 4) shows very little head motion - no volume exceeds the 0.5 mm threshold, so no censoring was needed, and motion is not visibly correlated to the story/math blocks. With a stricter threshold, the frames around 60s would probably need to be scrubbed.

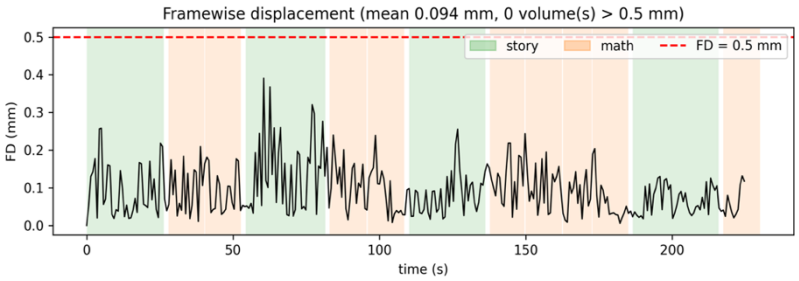


Figure 4 Framewise displacement between subsequent volumes. The 0.5mm FD threshold is displayed as a red dotted line, and the math/story blocks are displayed as shaded areas.

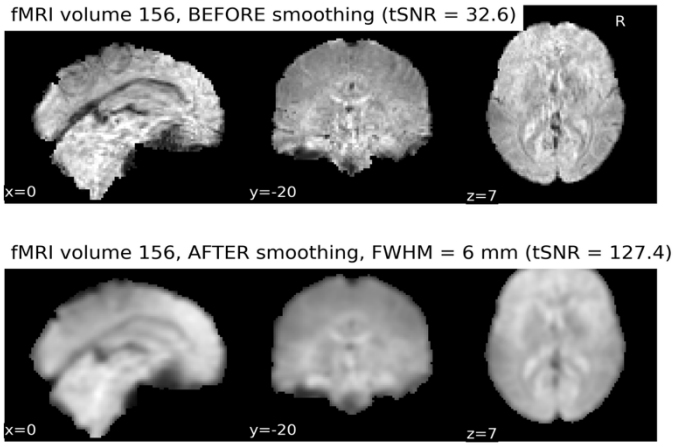


Figure 5 Middle volume of fMRI time series before and after gaussian smoothing with FWHM = 6mm.

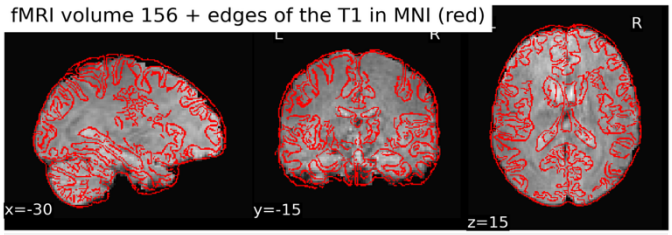


Figure 6 Anatomical MNI-registered T1w (one slice) overlaid on fMRI volume for alignment assessment.

Smoothing. A 3D Gaussian with a 6mm FWHM (three times the 2mm voxel size) was applied inside the brain mask. This increases the signal-to-noise ratio, since noise is mostly independent between voxels while activations span several voxels, and brings the data closer to the assumptions of voxel-wise statistics (Gaussian random fields). The cost is spatial specificity: small activations are diluted and grey-matter signal spreads into WM and CSF. 6mm was chosen since the data is only for a single subject (8mm is standard in group studies).

Anatomical–functional alignment. Figure 6 shows the MNI-registered T1w overlaid on a functional volume. The T1 edges follow the ventricles, the insula and most of the cortical folding. Minor discrepancies remain in the parietal cortex and at the base of the brain (inferior temporal lobes, cerebellum), potentially due to EPI signal drop due to the ear canal and the permissive BET threshold which captures some extra tissue. As a test, we let FLIRT register the mean fMRI to the T1: the residual displacement is only about half a voxel. Therefore, the coregistration of the fMRI to the T1 can be skipped. The HCP fMRI is already non-linearly normalised on the MNI 2 mm grid, and its internal structures line up with the non-linearly registered T1.

1.3 GLM

Design matrix. The event file describes two levels: the blocks story and math (~12–30 s each, alternating without rest), and the sub-events inside them (present, question, response, and a short auditory cue before each block). Our main model is a block design (story vs. math), which answers the project question with the most power and is the design used by Binder et al. (2011) [1]. It has two boxcar regressors (story, math) convolved with the canonical double-gamma HRF, plus nuisance regressors: the 6 motion parameters, 4 cosine drift terms (high-pass 0.01 Hz) and a constant (13 columns, Fig. 7). Story and math are strongly anti-correlated (r = −0.94): blocks alternate without rest and cover 94.5% of the run (213 of 225.4 s), so story + math ≈ constant, and both are almost collinear with the intercept. Their individual betas mean "versus the ~12 s of cues/gaps", not "versus rest". They mostly capture what is common to both tasks (auditory input, attention, button press). The story−math contrast, however, lies along the direction of greatest variation of the design (the alternation itself) and is well estimated. This is why the task was designed with matched block durations. The fourth drift

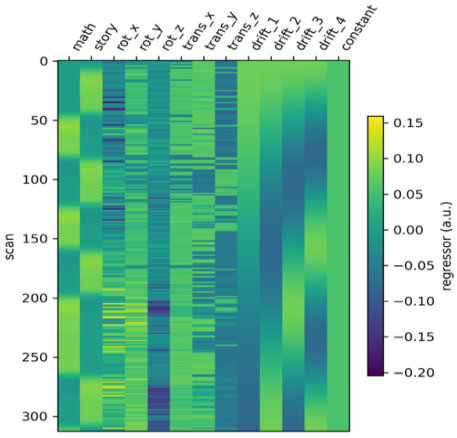


Figure 7 Design matrix for Block design experiment.

term (period ~113 s) correlates moderately with the task (r = 0.34 with story, −0.41 with math), because the four consecutive math blocks (138–187 s) add low-frequency content. A lower cut-off would reduce this, at the cost of less drift removal. Motion parameters are weakly correlated with the task (|r| ≤ 0.25), so they remove little task signal. They correlate more with the slow drifts (|r| up to 0.62) and, among themselves, rot_y and trans_x (r = 0.75).

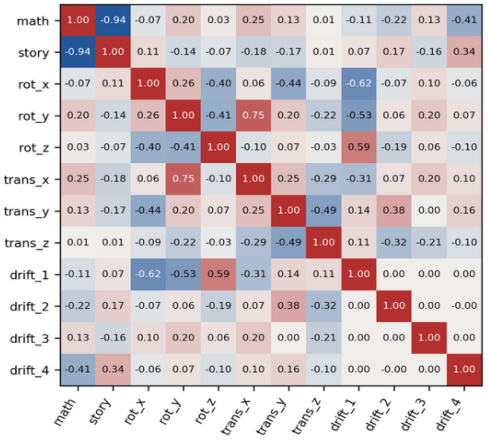


Figure 8 Correlation matrix of the regressors. Red values indicate a positive correlation, while blue values indicate a negative correlation. Grey, pale boxes indicate little to no correlation.

GLM and single-regressor maps. With one subject and one run, only a first-level analysis is possible. The GLM was fitted voxel by voxel on the preprocessed data. Maps are two-sided z-scores, FDR-corrected (q < 0.05), keeping clusters of at least 10 voxels. Each regressor is tested against the implicit baseline, i.e. the 12 s of the run that are covered by no block. As expected from the collinearity, the story and math maps are almost empty (145 and 1137 voxels, all negative); they are not interpretable alone (Figure 9).

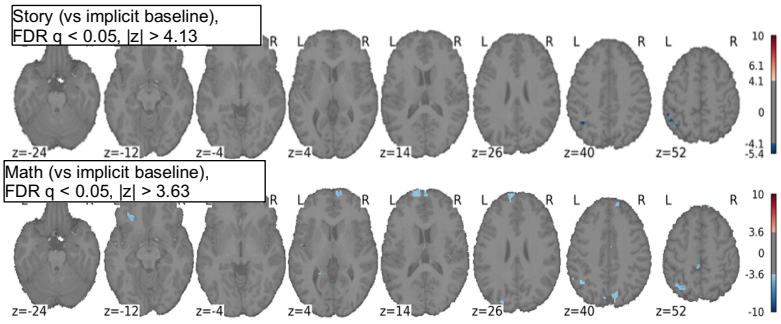


Figure 9 Activation maps for 'story' and 'math' against baseline (both FDR-adjusted).

Story vs. math contrast. Contrast vector **c = [−1, +1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]**. Unlike the single-regressor maps, this contrast is very strong (peak |z| ≈ 10; Figure 10). About one third of the mask is significant, so the FDR threshold drops to |z| > 2.42. The Bonferroni map (|z| > 5.19, ~12 000 voxels) controls the family-wise error rate and keeps only the core of each region, which makes regions easier to name.

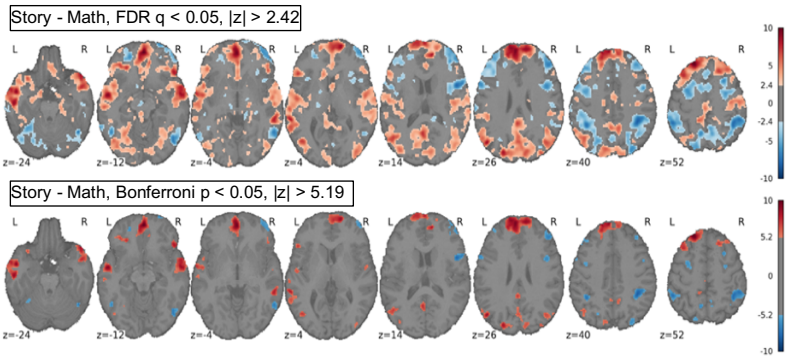


Figure 10 story − math maps, FDR (top) and Bonferroni (bottom) corrections, axial slices. Red: story > math, blue: math > story.

Regions and networks. Region names were read off the maps using the Harvard-Oxford atlas (FSL). *Story* > *math*: medial prefrontal cortex and left superior frontal gyrus, bilateral anterior temporal lobes, superior temporal sulcus, angular gyrus, posterior cingulate, precuneus, right posterior cerebellum. *Math* > *story*: bilateral intraparietal sulcus, right precentral, middle frontal gyrus, right lateral prefrontal cortex, inferior temporal cortex, occipito-temporal cortex. In terms of the resting-state networks of Yeo et al. (2011) [2], the *story* > *math* regions correspond to the semantic / default-mode network. The default-mode network is typically engaged by internally directed, integrative processing such as understanding a narrative, retrieving semantic knowledge and following a situation over time, which is what the story blocks require. The *math* > *story* regions correspond to the fronto-parietal network of numerical processing, attention and working memory, which is required for math problems (keeping operands in mind, computing). The coverage map we obtained can be found in the appendix (Annex 2). The ventral ATL or Wernicke's area do not show the expected signal in the *story* > *math* contrast map, as very few voxels pass the threshold. This could be explained by the fact the contrast removes shared attributes between the blocks, and that math problems are also spoken sentences that must be understood and answered. The subtraction thus cancels shared auditory, phonological and syntactic processing, and part of the semantic processing (number words have meaning), so regions engaged by both tasks are weakened even if they are semantic. Other contributing factors are the low power of a single subject with a single 3.8 min run and the 6 mm smoothing (which mixes the thin ventral ATL cortex with neighboring low-signal voxels).

1.4 Theoretical questions

1a) Use an F-test on the two task regressors, with contrast matrix [1 0 0 …; 0 1 0 …]. It tests whether at least one beta differs from zero. The F-test carries no sign or task information, so the single-regressor t-maps (story > baseline, math > baseline) are needed to see which task drives each region and in which direction.
1b) Model the question phase and answer phase as separate regressors in each task (story-question, story-answer, math-question, math-answer). The contrast question-answer is [1 −1 1 −1 0 0 …]. Positive values (question > answer) mark regions more engaged during the question or problem presentation (mostly auditory and comprehension processing). Negative values (answer > question) mark regions more engaged during the response (decision, motor areas).
1c) Add a regressor for the preparation period (just before the task stimulus) next to the task regressor. The contrast task−preparation is [1 −1 0 0 …]. Positive values mark areas more involved in performing the task, negative values areas more involved in preparing for it. The two regressors may be strongly correlated, though, so the correlation matrix should be checked.
2) Use an F-test comparing models of varying complexity: the full model against a reduced model without the story regressor. A significant result means that story adds explanatory power beyond the other regressors. The test must also be corrected for multiple comparisons across voxels (FDR or Bonferroni).
3) H0: the story−math effect averaged over both runs is zero ($c \cdot \beta = 0$ with c = [−1 +1 0 0 …]). H1: it differs from zero. The second run should be preprocessed exactly like the first. Runs should then be concatenated into a single GLM, with separate drift terms, motion confounds and baselines, but shared story and math regressors, and tested on the same contrast. Alternatively, each run could be fitted separately and the contrast estimates averaged, weighted by their variance (fixed effects).

2. MVPA (K-Means)

Number of clusters. Each timepoint (fMRI volume) is one sample and each brain voxel is one feature. The clustering groups timepoints whose whole-brain activity pattern is similar, so each cluster is a recurring brain "state", and its centroid (the mean pattern of its timepoints) is a spatial map. The mean value of each voxel over time of the preprocessed data is removed and the values are normalised by their standard deviation. K-means was run with the Euclidean distance from *sklearn.cluster.KMeans*. We chose K=10, the largest K before the stability drops and just before an inflection point in the elbow plot (Annex 3). It gives the finest partition that is still reproducible and yields the 10 clusters to be analysed later. K ≤ 5 is more stable but too coarse: it merges the task states with the slow fluctuations of the BOLD signal.

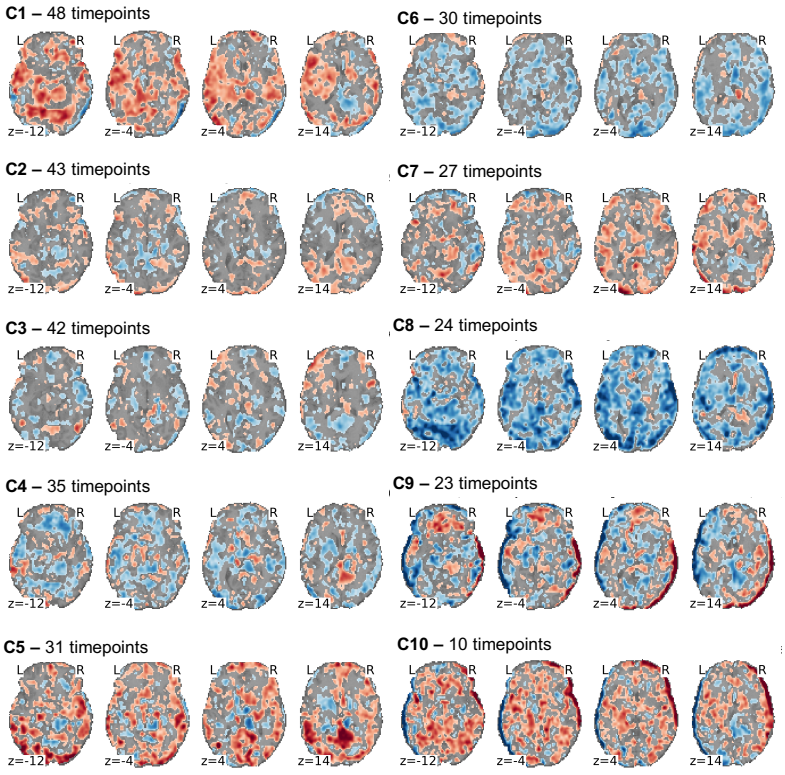


Figure 11 Spatial maps of K=10 cluster centroids overlaid on T1w axial slices.

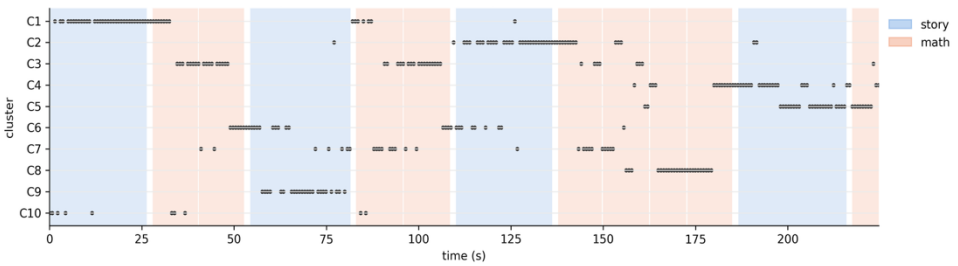


Figure 12 Binary time course. Each cluster timepoint is plotted over the experiment block design.

Spatial maps. Each centroid is shown in axial view on the T1 registered to MNI in units of standard deviation of each voxel (z), in Figure 11. Positive (red): the voxel is above its mean during the timepoints of the cluster, negative (blue): below.

Relation with the GLM. Temporal K-means finds the frontoparietal *math* state (C3, spatial correlation of -0.71 with *story-math* map and 0.47 with *math* regressor) and default-mode *story* states (C1 and C2, positive correlation with *story-math* map and with *story* regressor). However, they only explain part of the run, and the other clusters are dominated by drift, global signal and motion. The GLM separates these sources because it models them explicitly and tests the task with a predefined HRF-convolved model. The *math* C3 centroid is positive in the intraparietal sulcus and lateral prefrontal cortex (frontoparietal and dorsal attention networks) and negative in the posterior cingulate and precuneus (default mode). C1 covers the first *story* block and C2 the third one. C2 is highest in the default mode network and C1 is positive in most networks (limbic, somatomotor, default mode, visual), closer to a global increase of the signal at the start of the run. Each one covers a single block and are affected by the slow drifts (a story block at the start and one in the middle of the run are not similar enough to fall in the same cluster; Figure 12). The other clusters correlate with nuisance regressors: C9 and C10 capture motion, with only moderate correlation with the task, C7 and C8 capture the whole-brain global signal due to breathing, arousal and motion, and C4, C5, C6 capture segments at the end of the run, drift and motion.

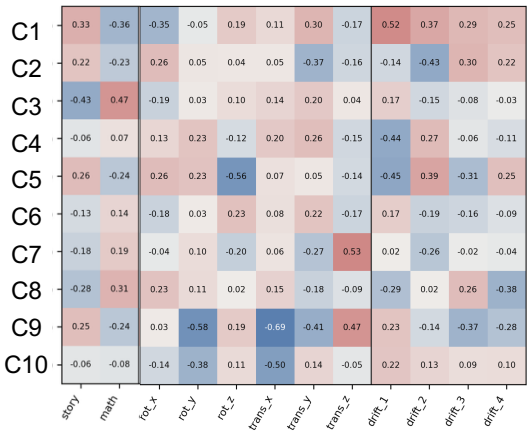


Figure 13: Correlation matrix of 10 clusters with GLM regressors. Red: positive correlation, Blue: negative correlation.

2.4 Theoretical questions

1) This signal has 313 timepoints and about 234 000 voxels/timepoint, so a centroid is a spatial map of about 234 000 values. With one voxel as one sample, we obtain 234 000 × 313 values. A centroid becomes a temporal profile of 313 values, the mean timecourse of a group of voxels, and the labels mapped back onto the brain give the spatial map. The clusters would then be functional networks or parcels and not full brain states. The problem is better conditioned, with many samples and few features, but slower, and each voxel's signal must be normalised so that the clusters follow the temporal shape and not the amplitude.
2) The comparison is partly appropriate. Both describe the BOLD signal with a few temporal patterns linked to brain regions, so checking whether cluster time courses match the task regressors makes sense. But the patterns are not of the same kind - A GLM regressor is a smooth, HRF-convolved predefined time course, while K-means is binary and data-derived. A cluster that captures a task state correlates only moderately with the regressor, with $|r| \leq 0.5$ here. The GLM is model-driven and univariate with known timing, a standard HRF and Gaussian noise, and tests each voxel independently. K-means is data-driven and multivariate, assumes compact clusters and groups timepoints by whatever varies most, so in our run most clusters follow slow drift, global signal or motion. In an experiment with a clear hypothesis and timing, such as this one, a GLM would be preferable, because it is more powerful and allows inference. We would use K-means when the timing is unknown, e.g. at rest, or to look for unexpected effects and artefacts in the data. The two complement each other as K-means shows what the GLM may miss and the GLM gives meaning to the K-means patterns.

Contributions of each group member

- Marbach Valentin: GLM, MVPA code, analysis part 1 and 2
- Schaffhauser Kilian: Functional preprocessing, theoreticals part 1 and 2
- Martignoni Matteo: Structural preprocessing, analysis part 1 and 2, writing the report

AI use statement

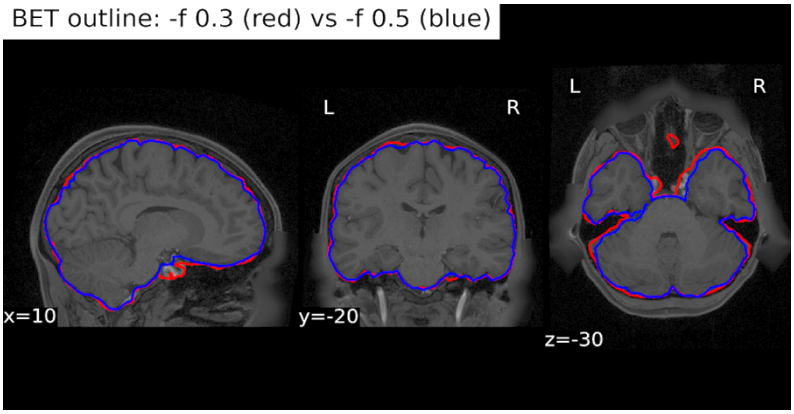
We used Claude (Anthropic) as a coding assistant. It substantially contributed to the code of Part 1.3 (design matrices, GLM, atlas analysis) and Part 2 (K-means pipeline, choice of K, correlations), proposed additional checks for Part 1.1 (BET threshold comparison, non-linear registration) and assisted Part 1.2 coding. The analysis of results, the theoretical answers, the report and the video presentation were written by the group without AI. We reviewed all AI-assisted code and comments step by step, re-ran all the code ourselves, corrected parts that did not match the outputs, and made the analysis choices.

References

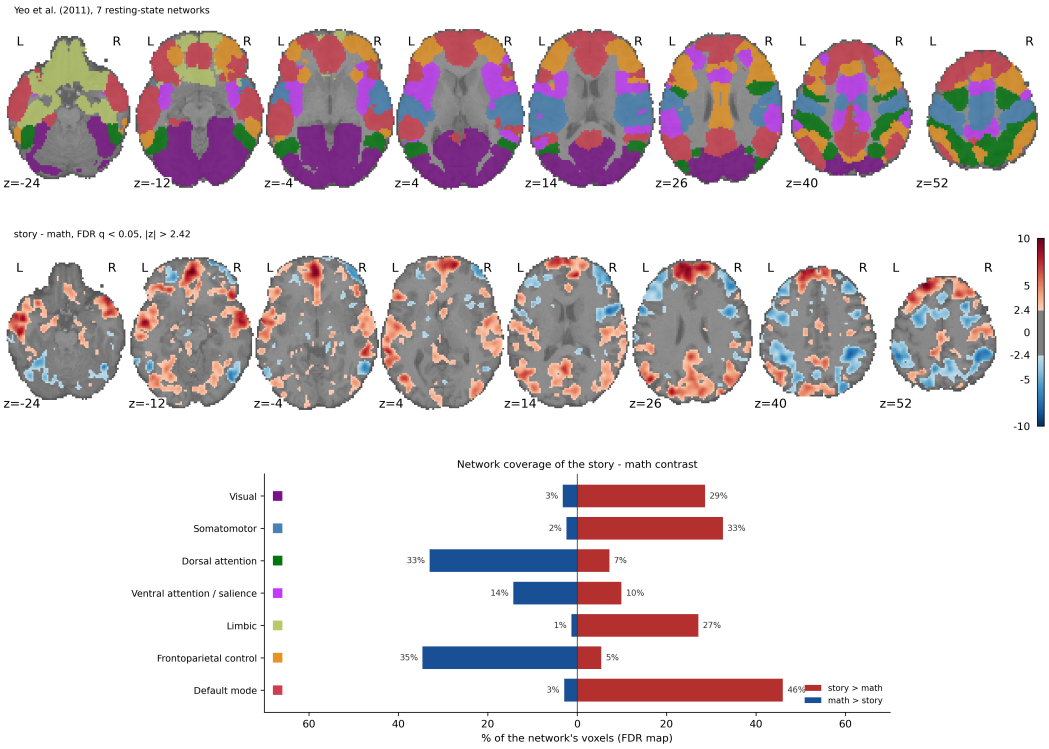
[1] Binder, J. R., et al. (2011). Mapping anterior temporal lobe language areas with fMRI: a multicenter normative study. *NeuroImage*, 54(2), 1465–1475.

[2] Yeo, B. T. T., et al. (2011). The organization of the human cerebral cortex estimated by intrinsic functional connectivity. *Journal of Neurophysiology*.

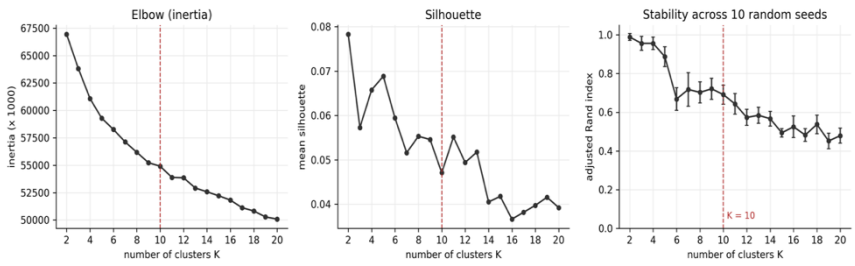
Appendix



Annex 1 Comparison of 0.3 and 0.5 threshold for the FSL Brain Extraction Tool.



Annex 2 Resting-state networks from Yeo et al (2011) [2] with the story-math contrast activation map and the overlap between the activations and the network map.



Annex 3 Elbow plot, silhouette score and stability of initialisation for K-means, computed for different numbers of clusters. Red vertical at K=10 clusters.